Topic 5 — Habitat (वास स्थान), Flora (वनस्पति) & Fauna (प्राणीजात)

Covers syllabus (click to expand)

Habitat | Habitat Conservation | Flora and Fauna

Sources baked in: NCERT Class 12 Biology, BSI/ZSI, MoEFCC, NUHHP 2007, MISHTI (Budget 2023–24), UPPCS PYQs 2018–2025 **Weight:** ★★ Medium–High — habitat definitions, MISHTI/NUHHP, BSI/ZSI, UP fauna **Last verified:** August 2026

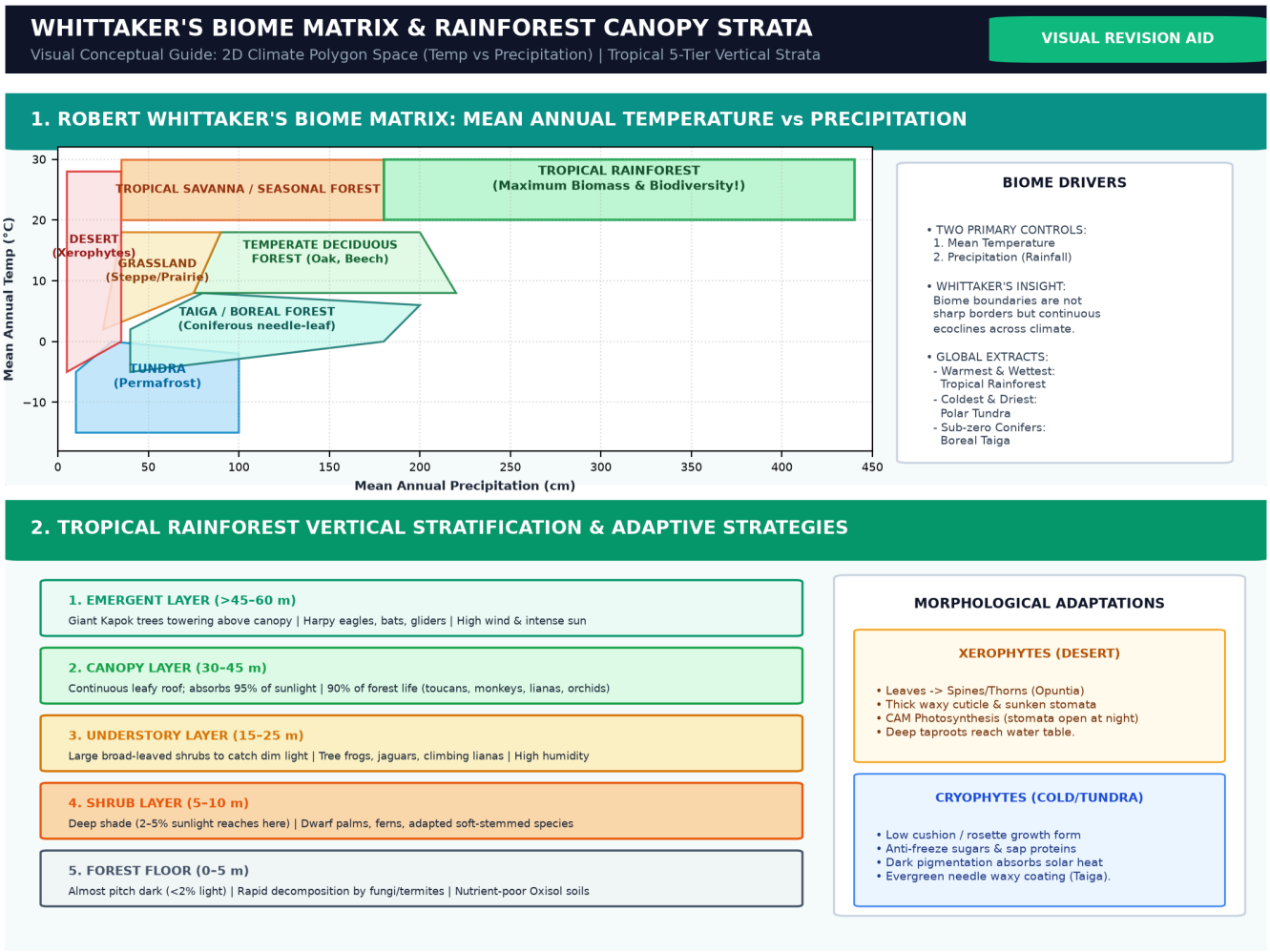
Current Affairs (this topic)

- Habitat-loss and corridor news often refresh Terai (तराई) / Himalaya (हिमालय) colour for UP–UK.
- Do not swap endemic vs endangered wording when CA names a new Red List update.

5.1 Habitat

Identity: Place where species lives | Terrestrial / aquatic / transitional | Adaptation-based types

- A **habitat** is the natural place where a plant or animal normally lives and finds food, shelter, and breeding conditions.
- NCERT treats habitat as the **address** of a species. It answers the question "where does it live?"
- A **habitat** is not the same as a **niche**. The niche is the species **role** in the ecosystem.
- A **habitat** is also not the same as an **ecosystem**. An ecosystem includes all interactions between living and non-living parts of an area.
- **Terrestrial habitats** exist on land. Organisms that live on land are called **terrestrial habitants**.
- **Aquatic habitats** exist in freshwater or marine water. Rivers are lotic and lakes are lentic aquatic habitats.
- **Transitional habitats** such as wetlands, mangroves, and estuaries are neither purely terrestrial nor purely aquatic.
- **Abiotic factors** in a habitat include temperature, moisture, light, soil, pH, and salinity.
- **Biotic factors** include food availability, predators, competitors, and symbiotic partners.
- A **microhabitat** is a small unit within a larger habitat, such as bark crevices or the underside of a leaf.
- A **macrohabitat** is a landscape-scale habitat unit such as an entire forest, lake, or desert tract.
- A **biotope** is a European ecology (पारिस्थितिकी) term for an area with relatively uniform environmental conditions. Notes often treat it as similar to habitat.



Major habitat types (classification)

Type	Meaning	India example
Terrestrial	Land-based	Terai sal forest, Thar (थार) desert
Aquatic freshwater	Ponds, lakes, rivers	Ganga, Loktak
Marine	Oceans and seas	Arabian Sea coral/ fisheries belt
Wetland	Seasonally or permanently waterlogged	Keoladeo (केवलादेव), UP Ramsar (रामसर) sites
Mangrove / coastal	Tidal saline (लवणीय) coast	Sundarbans (सुंदरबन)
Grassland / savanna (सवाना)	Grass-dominated open tract	Dudhwa grasslands
Alpine / montane	High-altitude cold zone	Himalayan snow leopard belt
Desert / arid	Low rainfall extreme temperature	Thar
Arboreal	Tree-dwelling	Lion-tailed macaque
Fossorial	Burrowing underground	Indian pangolin
Cursorial	Open plain runners	Blackbuck

Type	Meaning	India example
Xericole	Desert-adapted	Camel, desert fox
Amphibious	Both land and water phases	Crocodile, bullfrog

- India's richest natural wildlife habitats in UP lie in the **Terai belt** at Dudhwa, Kishanpur, and Pilibhit.
- The **Bushmen** live in the Kalahari Desert.
- The **Eskimo** live in Arctic Canada.
- The **Pygmies** live in the Congo Basin.
- The **Masai** pastoralists live in **East Africa** in Kenya and Tanzania.

PYQ — UPPCS Prelims 2024, Q62

Plants and animals that live on land are called:

A. Abiotic

B. Aquatic habitants

C. Terrestrial habitants

D. Biotic

Show answer

Ans: C

PYQ — UPPCS Prelims 2025, Q23

Which of the following pairs is/are NOT correctly matched?

(Tribe) — (Habitat)

1. Bushman — Kalahari Desert
2. Eskimo — Northern Canada
3. Pygmy — Congo Basin
4. Masai — West Africa

A. Only 1, 2 and 3

B. Only 4

C. Only 3 and 4

D. Only 1

Show answer

Ans: B — Masai = **East Africa**, not West Africa.

5.2 Habitat Conservation

Identity: Protect the place first | Loss / fragmentation / degradation | MISHTI + NUHHP

- **Habitat conservation** protects the natural dwelling place of species so ecological processes can continue.
- Without habitat protection, species-level measures such as captive breeding alone cannot succeed long term.
- **Habitat loss** completely removes an area's ability to support native species.
- **Habitat fragmentation** breaks a large habitat into small isolated patches and blocks movement and gene flow.
- **Habitat degradation** reduces habitat quality without removing it entirely, such as river pollution affecting dolphin habitat.
- **In-situ conservation** protects species inside their natural habitat. National parks and wildlife sanctuaries are examples.
- **Wildlife corridors** link fragmented habitats. The **Terai Arc Landscape** connects Dudhwa with Katarniaghat across the Nepal border.
- **Habitat restoration** rebuilds damaged ecosystems through afforestation, wetland revival, and mangrove planting.

Key schemes and policies

Scheme / policy	Year	Ministry	Fact
MISHTI	Budget **2023–24**	MoEFCC	Mangrove shoreline habitat restoration + community income
NUHHP	**2007**	MoHUA	Urban housing and sustainable urban habitat
JNNURM	**2005**	MoHUA	Urban infrastructure renewal
AMRUT	**2015**	MoHUA	Urban water, sewerage, green spaces
Jal Jeevan Mission	**2019**	Jal Shakti (शक्ति)	Rural tap water connectivity
CAMPA	2009 / Act 2016	MoEFCC	Compensatory afforestation for diverted forest land
Green India Mission	2014	MoEFCC	Forest cover (वन आवरण) and habitat quality improvement under NAPCC

- **MISHTI** stands for **Mangrove Initiative for Shoreline Habitats and Tangible Income**.
- MISHTI targets coastal mangrove habitats for shoreline protection and blue carbon (नीला कार्बन) restoration.
- **NUHHP** addresses urban habitat quality through housing and green-space planning in cities.

PYQ — UPPCS Prelims 2025, Q93

With reference to Mangrove Initiative for Shoreline Habitats and Tangible Income (MISHTI), which of the following statements is/are correct?

1. It was initiated in 2019.
2. It aims to restore mangrove forests.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

Show answer

Ans: A — Only statement 2. Launch = **2023–24** Budget.

PYQ — UPPCS Prelims 2025, Q72

Consider the following and arrange them in the correct chronological order:

1. Atal Mission for Rejuvenation and Urban Transformation (AMRUT)
2. Jawaharlal Nehru (जवाहरलाल नेहरू) National Urban Renewal Mission (JNNURM)
3. Jal Jeevan Mission
4. National Urban Housing and Habitat Policy

A. 2, 4, 1, 3

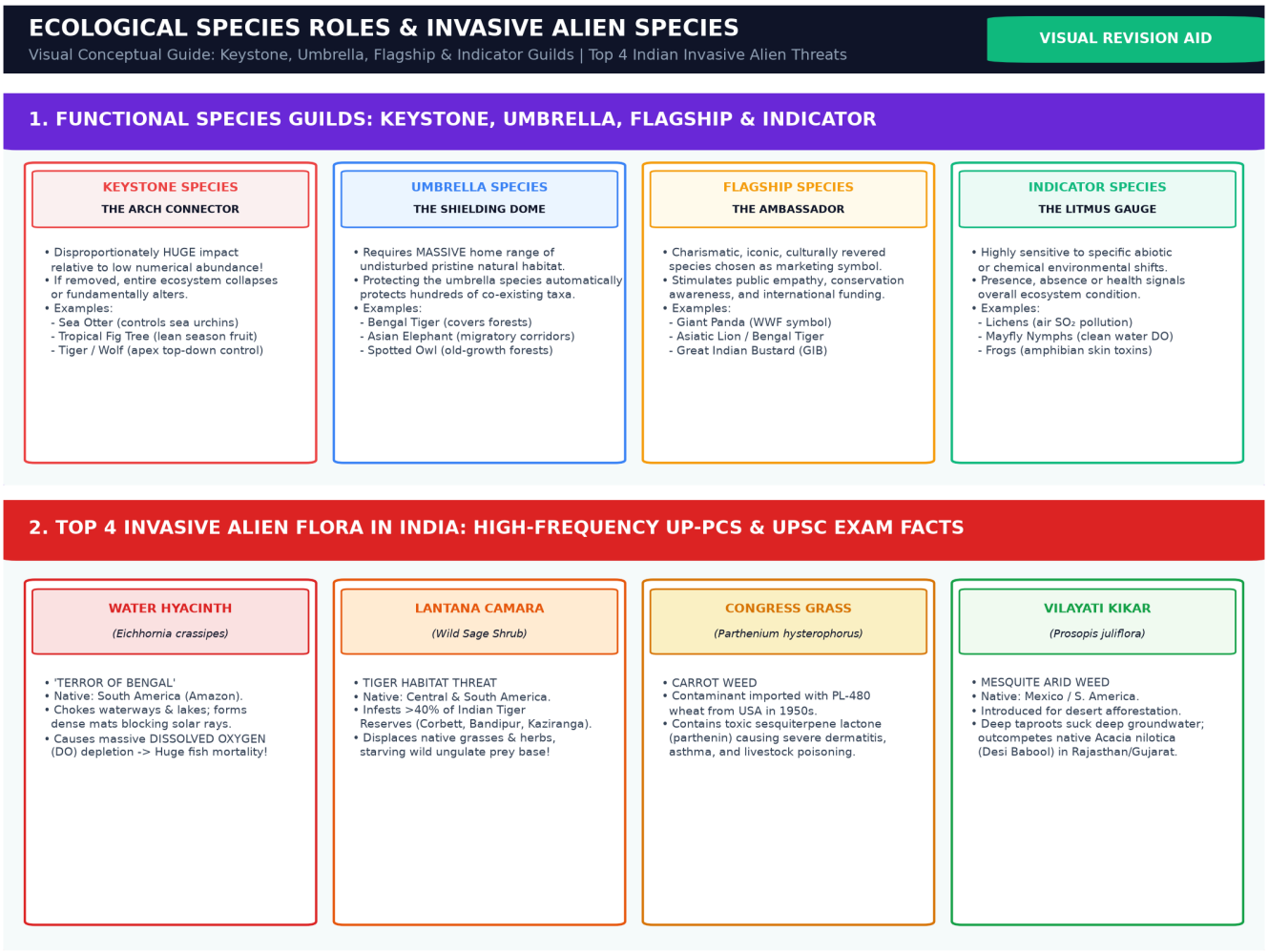
B. 4, 2, 1, 3

C. 2, 4, 3, 1

D. 4, 2, 3, 1

Show answer

Ans: A — JNNURM 2005 → NUHHP 2007 → AMRUT 2015 → JJM 2019.



5.3 Flora and Fauna

Identity: Biotic inhabitants of habitats | BSI + ZSI | India megadiversity stats

- **Flora** is the collective **plant** life of a particular region, habitat, or geological period.
- **Fauna** is the collective **animal** life of a particular region, habitat, or geological period.
- Flora includes trees, shrubs, herbs, grasses, ferns, and algae. It is **not** limited to flowering plants.
- Fauna includes invertebrates and vertebrates across insects, fish, birds, and mammals.
- Together, flora and fauna form the **biotic content** of a habitat.
- India has about **45,000** plant species and **91,000+** animal species on roughly **2.4%** of Earth's land area.
- The **Botanical Survey of India (BSI)** (भारतीय वनस्पति सर्वेक्षण) was established in **1890**. Its headquarters is in **Kolkata**. It surveys Indian **plants**.
- The **Zoological Survey of India (ZSI)** (भारतीय प्राणी सर्वेक्षण) was established in **1916**. Its headquarters is in **Kolkata**. It surveys Indian **animals** and publishes the *Fauna of India* series.
- **Adolf Engler** mapped the world's major **floral regions**.
- **C.O. Sauer** introduced the concept of **planting hearth**, the region where domesticated plants first originated.
- The **Puszta (पुस्ज़टा) grassland** is associated with **Hungary**.
- The **cryophyte steppe (स्टेपी)** is associated with **Arctic tundra (टुंड्रा)** vegetation.

- **Red Sanders (*Pterocarpus santalinus*)** is a South India dry-deciduous endemic (Palkonda–Seshachalam / southern Eastern Ghats (पूर्वी घाट)) — **not** a tropical rainforest tree.
- **Taxus** occurs naturally in the Himalaya, is Red Data–listed, and yields **taxol** (anti-cancer; older stems also cite Parkinson's).
- Himalayan associates include **oak** and **rhododendron**; **sandalwood** is South Indian deciduous — not a Himalayan natural associate.
- Desert flora often shows hard/waxy leaves, tiny leaves, or thorns to cut water loss.

UP flora and fauna facts

Category	UP example	Note
State animal	**Barasingha** (swamp deer)	Terai grassland specialist
State bird	**Sarus crane**	Wetland and farmland mosaic
Dominant Terai flora	**Sal** (*Shorea robusta*)	Dudhwa forest tree
Aquatic fauna	**Gangetic dolphin**	Ganga–Yamuna stretch
Flagship habitat fauna	Tiger, rhino, elephant	Dudhwa and Pilibhit
Invasive flora	Parthenium, Lantana, water hyacinth	Degrade native habitats

- UP has **no coastline**, so mangrove flora is **not** found in the state.

> **Logic:** BSI = **plants**. ZSI = **animals**. Both HQ **Kolkata**. Trap — BSI documents animal diversity is **false**.

PYQ — UPPCS Prelims 2024, Q29

Match List-I with List-II:

List-I: A. Planting Hearth B. Cryophyte Steppe C. World Floral Regions D. Puszta Grassland List-II: 1. Hungary 2. Adolf Engler 3. C.O. Sauer 4. Arctic Tundra

A. 3 4 2 1

B. 2 1 4 3

C. 3 4 1 2

D. 2 3 4 1

Show answer

Ans: A — Sauer–3, Arctic–4, Engler–2, Hungary–1.